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1. Explain the OSI (Open system interconnection) model in detail. Describe all seven layers of the OSI model, their functions and the role each layer plays in data communication.

1. Physical layer (layer 1): Deals with transmission of raw bits over a physical medium. Define hardware elements like Cables, Switches, voltages levels. Converts data into electrical / optical signals.

2. Data link layer (layer 2): Responsible for node-to-node data transfer. Detects and corrects errors. Uses MAC address for communication. Divides data into frames.

3. Network layer (layer 3): Handles routing and logical addressing. Determines the best path for data transmission. Uses IP address. Example Routers, IP protocol.

4. Transport layer (layer 4): Ensure reliable data transfer. Provides error recovery and flow control. Segments data into smaller units. Example: TCP, UDP.

5. Session layer (layer 5): Manages session between applications. Establishes, maintains and terminates communication. Synchronizes data exchanges.

6. presentation layer (layer 6) : translates data into a readable format. Handles encryption and decryption. perform data compression.

7. Application layer (layer 7) : closest to the user provides network services to application.
Examples : HTTP, FTP, SMTP

2. Explain the TCP/IP model in details. Describe all four layers of the TCP/IP model, their function, and commonly used protocols at each layer. How does the TCP/IP model support communication over the Internet.

— The TCP/IP model is the practical model used for Internet communication. It consists of four layers and is simpler than the OSI model.

1. Application layer : Combines OSI Application, presentation, and session layers.
provides services to user applications.
protocols :
• HTTP
• FTP
• SMTP
• DNS

2. Transport layer : provides end-to-end communication, ensure reliability and error handling.
protocols : TCP (reliable, connection-oriented)
• UDP (fast, connectionless)

3. Internet layer: handles logical addressing and routing. Sends packets across network.
 protocols: IP (Internet protocol)
 ICMP (error reporting)
 ARP (address resolution)

4. Network Access layer: Combines OSI physical and data link layers. Deals with hardware and transmission.
 protocols: Ethernet, Wi-Fi

Advantages: Simple and Scalable, Supports global communication, Reliable and efficient

Thus, the TCP/IP model forms the backbone of the Internet.

Breaks data into packets, assign IP address for routing, uses TCP/UDP for delivery. Ensures packets reach destination correctly. Reassembles data at receiver side.

3. Discuss the similarities and differences between the OSI Model and TCP/IP model with respect to architecture, functionality, protocol mapping and real-world application. Why is the TCP/IP model more widely used in practical networking despite the OSI model being more descriptive?

Similarities:

- Both are layered network models.

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- Both divide Communication into logical steps
 - Both supports data transmission across network
 - Both have transport and network functionalities
 - Both ensure interoperability between system.

Differences

Aspect	OSI model	TCP/IP model
layers	7 layers	4 layers
Developed by	ISO	DARPA
Nature	Conceptual model	practical model
layer separation	Strict	flexible
protocol dependency	Independent	protocol-based
usage	Educational	Real-world networking

Protocol Mapping: OSI application + presentation + Session → TCP/IP application

- OSI Transport - TCP/IP Transport
- OSI Network - TCP/IP Internet
- OSI Data link + Physical → TCP/IP networking

Real-world application: OSI is used for learning and troubleshooting.

- TCP/IP is used in Internet, LAN, WAN, Cloud Computing